

Air Compressor FAQs

Our customers ask us a ton of questions. We value each one as an opportunity to help you meet your compressed air needs and achieve your bottom line goals.

To help you benefit from the wisdom of your predecessors, we've put together this list of frequently asked questions about air compressors and related equipment. Use the link list at right to navigate to the section you want quickly and easily.

Below are the most frequently asked questions regarding compressors:

Air Quality FAQ's

Q: My compressor runs hot. What should I do ?

A: Elevated compressor temperature may be caused by restricted air, water or all flow, a low oil level, or excessive ambient temperature. Check the water flow, pressure and quality, and clean the heat exchanger as needed. Check the compressor oil level and add oil as required. Remove any flow restrictions and replace parts in necessary. Make sure your compressor and compressor room are properly ventilated. Check with the manufacturer to determine the maximum operating temperature.

Q: What are the most common problems caused by water in compressed air systems ?

A: Washing away of required lubrication, increased maintenance, sluggish pneumatic equipment, rust instrumentation clogs, paint spotting, airline freeze-up and dramatically shortened tool life.

Q: Why am I getting water in my air lines ?

A: Condensation in the tank and air lines is a natural occurrence when compressing air. To prevent water in the lines, drain the air receiver frequently to eliminate the condensate. Consider adding an automatic drain to eliminate the manual process of draining the tank. Add a moisture separator to your air line and drain on a regular basis. If you have an air dryer, water in the lines may be a sign that you need to repair or replace your dryer. Have the dryer serviced to see if that eliminates the water problem.

Q: What is the hidden cost of bad air quality ?

A: Bad Air = Repair Cost + Replacement Cost + Lost Production.

Q: Why am I getting low air pressure at the point of use ?

A: Low pressure can be caused by a number of factors. You may have air leaks in the distribution piping or clogged filter elements. The pressure at the compressor discharge may not be sufficient to overcome system pressure drop. Check the lines, connections and valves for leaks. Clean or replace filter elements. Raise the compressor discharge pressure as a last resort.

Q: The pressure at compressor discharge is low. What should I do ?

A: For systems with modulating load controls, the air capacity control system may not be adjusted properly. Follow the manufacturer's recommendation for adjustment of the modulation system. In reciprocating compressors, another cause may be worn or broken valves or the wrong setting on your air pressure switch. Check the valves and repair or replace as required. Follow the manufacturer's recommendations for setting the air pressure switch. If none of the above solves the problem, the low-pressure problem may be caused by an undersized compressor. You may require additional capacity (CFM) to maintain the desired system pressure.

Q: What can I do to prevent oil in the air lines ?

A: All lubricated compressors send oil downstream. The quantity will depend on the type of compressor. In a rotary screw compressor, oil in the lines may be due to faulty separation equipment or a compressor oil level that's too high. Check the air/oil separation system and change the separator element if necessary. Also follow the manufacturer's recommendations for the proper oil level. In a reciprocating compressor, the force feed lubricator may be set at an excessive level. Follow the manufacturer's recommendation for proper feed rate. If oil carryover continues to be a problem, you may need to add a coalescing filter to your system.

Q: Why am I getting dirt or rust in the air lines ?

A: As compressed air lines age, dirt can accumulate in the air lines and rust can be caused by water carried over to the piping system. To prevent damage to your equipment, install filters at the point of use. When replacing or adding piping, consider the use of copper or extruded aluminum piping which will not rust. Adding a dryer to remove water before the air enters the piping system will reduce rust problems.

Air Compressor Costs FAQs

Q: What is the cost of pressure drops ?

A: Avoid costly pressure drops and save money. NOTE: A 10 hp compressor producing 35 cfm at 100 psig and connected to a straight run of 100 feet of 1" diameter pipe will result in a .27 psi pressure drop. Every 90° elbow will equal to an additional 25 feet of pipe. Four elbows, another loss of 1.12 psi. Each lost pound of pressure equals the loss of .5 percent horsepower. Horsepower equates to power; power equates to money.

Q: What is the cost of an air leak ?

A: Eliminate air leaks and save money. The cost of a 1.8" air leak at 100 psi (26 cubic feet per minute) represents INR 20 per hour or INR 2 Lacs for every 10,000 hours of operation. Fix the leak and save money.

Air Compressor Efficiency

Q: How can I improve the energy efficiency of my rotary screw compressor ?

A: The most efficient compressor is one that is fully loaded, providing the maximum flow per brake horsepower. When an application has long periods of little or no demand for compressed air, you should have a compressor that can shut itself off.

Q: Can I really recover energy from my compressed air system ?

A: As much as 80-93 percent of the electrical energy used by an industrial air compressor is converted into heat. In many cases, a properly designed heat recovery unit can recover anywhere from 50 to 90 percent of this available thermal energy and put it to useful work heating air or water. In the winter, you can supplement your building's heating system with temperature controlled, heated make-up air from your compressor. During the summer, you can isolate the waste heat from your building's comfort conditioning system. The amount of energy that you can save through recovery is a function of your current heating system efficiency.

Air Systems Equipment

Q: Can an air dryer really make much of a difference ?

A: Hot air holds more water. Every 20° F increase in temperature doubles the amount of moisture suspended in air. Every 20 F decrease in temperature reduces the amount of moisture suspended in air in half. An aftercooler or air dryer can eliminate this fluctuation. Air entering a 10hp compressor at 75° F and 75 percent relative humidity would produce 6.3 gallons of water per day. An air-cooled aftercooler with a separator would remove 4.3 gallons per day and a refrigerated air dryer would remove an additional 1.7 gallons. Only .3 gallons would be left within the 50,400 cubic foot of compressed air.

Q: How much water is generated by my air system ?

A: One 100 cfm compressor operating at 100 psig will generate 18 gallons per day at 90° F and 50 percent RH. About 60 percent of this moisture will be removed by the aftercooler. The remainder will pass into your compressed air system if a dryer is not installed.

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Q: Which filters and dryers do I need for my application in order to eliminate moisture and contaminants in my airlines ?

A: This depends on the application and required dewpoint. For most applications, a 35° F dewpoint is sufficient, which can be provided by a refrigerated air dryer. For outdoor applications, a desiccant dryer will be necessary to provide a -40° F dewpoint. Filters will provide various levels of contaminant removal and can be selected according to the application requirements.

Q: How many cubic feet does a receiver hold ?

A: Only stored air above the pressure being used is defined as useful free air. Majority of pneumatic tools and work processes only require 90 to 100 psi. Storing compressed air above this working pressure provides a surplus of useful air that can accommodate work demands. A 120-gallon receiver at 175 psi stored provides 193 cubic feet of compressed air.

Q: What size piping is needed in a properly designed system to minimize pressure drop ?

A: This would depend on the capacity (CFM) of your air compressors, the required pressure and the total length of piping in the system. By knowing those three factors, standard tables can be used to determine the size of pipe required to minimize pressure drop. It is also important to know the number of valves and elbows in the system, as each one counts as an equivalent length of straight pipe. Size piping for the maximum CFM required and include the current system capacity and possible future expansion in your piping plan.

Compressor System Maintenance FAQs

Q: What kind of maintenance should be done to my compressor ?

A: On oil-lubricated units, you should check the oil level regularly and change it as specified in the owner's manual. Also, change the oil filter as the manufacturer suggests. Clean or replace the air filter on a regular basis. On oil-less units, the air filter must be kept clean or debris will enter the compression chamber, resulting in high repair costs. Remember to drain condensate from the tank regularly.

Q: What are the main areas of a compressor package that need maintenance ?

A: Compressor, heat exchanger surface, air lubricant separators, lubricant, lubricant filters and the air inlet filter.

Q: How can I tell if my drain valves are operating properly ?

A: Routinely inspect condensate drains. Improper operation will exhaust compressed air to ambient or flood compressed air system with liquid water and oil. Clear tubing is recommended between electric drain and condensate collection point - a transparent tube provides easy inspection of drain operation.

EFFICIENT COMPRESSED AIR SYSTEMS

HOW TO SAVE ENERGY, REDUCE COSTS AND HELP THE ENVIRONMENT

COMPRESSED AIR TREATMENT

All compressed air needs some form of treatment to achieve the desired air quality. This guide investigates the various technologies available to treat compressed air and

SEPARATOR AND CONDENSATE DRAINAGE

The accumulation of condensed water in the system can lead to pressure drop. In lubricated compressors the oil vapour can condense with the water and form an emulsion. Additionally, wear particles and pipscale can amplify the problem by becoming trapped in valves and diaphragms. Water condensation can occur throughout a compressed air system and it is good practice to remove as much of this water as possible.

AFTER COOLERS

Nowadays, most compressors have an after-cooler as a standard fitting to perform the function of removing water, and some oil vapour. However, after coolers are also available as free standing units, in air blast or water cooled versions, for compressors not fitted with them. The after-cooler, through condensing, usually reduces the water content of the air by around 68% and uses approximately 2% of the total package power.

When air mains travel outside the compressor room, temperatures are typically lower and lead to condensation. As such, the air mains should slope to drain legs located at strategic intervals. Drain points should be constructed with a fall of not less than 1 meter in 100 meters of ring main pipe-work. The recommended distance between drain points is approximately 30 meters. If there is a need for branch lines, these should be installed at the top of the main to prevent water running into it from the main pipe. Branch line pipes should also be drained. In order to remove the water drain traps need to be fitted to the drain points.

In order to ensure drain traps perform efficiently, it is strongly recommended that simple wire type strainers be fitted prior to the drain trap to prevent particulate contamination of the trap. Occasional maintenance is also required. Oil or emulsion build-up needs to be removed regularly and if the build-up is heavy, drain trays, which have a blast action discharge, should be considered. Making wise drainage decisions can save up to 10% of the costs of compressed air.

DRYERS

Water in compressed air: There are 3 phases of water in compressed air being liquid water, aerosol (mist) and vapour (gas). The most noticeable and easily removable are water and aerosol which can be removed by high efficiency filtration together with refrigeration dryers. Water vapour is more difficult to remove and require the use of a desiccant dryer together with high efficiency filtration.

Removing water and aerosol can be satisfactory for general industry where corrosion damage is not an issue. If there are concerns about corrosion damage inside the compressed pipe work, together with ensuing rust and debris which may damage and contaminate downstream and products, then desiccant quality air should be considered.

The general measurement of air dryness is dewpoint, which can cause confusion when choosing the required dryer for the system, particularly in generally warm countries such as Australia. It should be noted that to stop corrosion (and rust, etc.) Air dryness should be 2% RH (Relative Humidity) or better. A dryness of 2% RH is equivalent to -30° C Pressure Dewpoint. Table 1 provides a comparison guideline for air dryness. It is also relatively easy to calculate how much more water (Liters) can be removed from a compressed air pipe work by alternative dryers.

Table 1. Pressure Dewpoint and Relative Humidity

Pressure Dewpoint	Relative Humidity
+10°C	54%
+3°C	34%
-20°C	5%
-30°C	2% (Limit of corrosion)
-40°C	0.7%
-70°C	0.016%

Typically atmospheric air passing through a compressor is one eighth its previous volume. Yet still contains the same amount of contaminants (water and particulate). Increasing pressure would normally cause moisture to condense out of the air, however, due to frictional heat the temperature of the air rises during the compression process and increases the compressed air's rises during the compression process and increases the compressed air's rises during the compression process and increases the compressed air's ability to hold water vapour. The water content of compressed air can be decreased using dryers. Dewpoint is the temperature below which water vapour will condense to liquid water at given conditions. Lowering the dew point effectively means the system can endure much lower temperatures before water droplets begin to condense. Essentially for every 11°C drop in compressed air temperature, the moisture holding capacity of air is reduced by 50%. Therefore, drying prevents liquid water forming downstream where it can contaminate or damage the system causing operating problems, costly maintenance, and repair expenses.

When selecting a dryer you need to determine the most cost-effective system suitable for your application as compressed air dryers vary in relation to their dew point, initial cost, and ongoing maintenance requirements. The factors to be considered include:

calculating the required dew point temperature, which needs to be below the lowest ambient temperature your compressed air system will be exposed too. Take into account the location of air lines i.e. Located in front of open doors or windows, throughout air conditioned or unheated areas, running underground or between buildings

Determining which type of dryers will produce the required dew point

Consider initial and operating costs. The lower the dew point, the more expensive the dryer is to purchase and operate

After selecting the type of drying system required, determine the actual conditions under which the dryer will be operating. This allows you to determine the correct size of the dryer.

- Max flow capacity (1/sec)
- Max acceptable pressure dew point (°C)
- Minimum inlet air pressure (kPa)
- Maximum and minimum inlet air temperature (°C)
- Maximum ambient or cooling water temperature (°C)
- Maximum allowable pressure drop (kPa)

There are five main types of dryers suitable for compressed air systems and each will perform differently and will be suited to different applications. Table 2 summarises the performance and approximate energy costs for the different types of dryers. More detailed descriptions of each dryer type follow.

Table 2. Typical additional costs for drying compressed air

Pressure Dewpoint	Dryer type	Added Energy Cost
+10°C	Deliquescent	1%
+3°C	Refrigeration	5%
-20°C	Membrane	28%
-20°C	Sorption	3-5%
-40°C	Desiccant Heatless	10-15%
-40°C	Desiccant Heated	8-12%
-70° C	Desiccant Heatless	21%

REFRIGERATION DRYERS (15°C TO 2°C DEWPOINT)

The refrigeration dryer cools the incoming air condensing moisture out. The dried air is then re-heated by the incoming air in the air-to-air exchanger. This method of drying is very popular as it produces dew points. Which are adequate for most duties in an energy efficient and reliable manner. The extra cost will typically be 5% over delivering standard after-cooled air, which will deliver pressure dewpoints of +3°C and remove an additional 28% of the initial water content. The use of a pre-filter prior to the dryer can not only protect the dryer from internal contamination (especially oil carryover) but also reduce the amount of liquid water (and particulate which can block drains) the dryers have to deal with and, therefore, increasing the dryers efficiency. Therefore, both oil lubricated and non-oil lubricated compressors require high efficiency filtration. In some climates winter ambient temperatures fall below the dryer dew point so air taken outside the buildings will cause water to condense. It is recommended that a condensate trap be fitted to the system where it enters the next building to prevent problems. Refrigeration dryers use well proven technologies that encounter few problems if properly installed and maintained. The types of problems that can affect the performance, and hence energy consumption include:

- internal contamination effects dewpoint - pre-filtration is recommended
- high compressor delivery temperatures
- high ambient temperatures
- poor installation/ ventilation
- faulty drain traps allowing liquids downstream of the dryer
- loss of refrigerant

DESICCANT DRYERS (-20°C TO -70°C DEWPOINT)

Desiccant dryers work by feeding compressed air through an alternating duty section while a non-duty section is used for regeneration. These units are designed to remove vapour phase moisture. There are several types of desiccant systems, the main two types being heated and heatless dryers. 'No Purge' low energy use desiccant dryers have recently been introduced to the market, these dryers work on similar principles to standard desiccant dryers, however, regeneration is carried out by the use of heated ambient air being drawn over the non-duty section under vacuum. All types usually require oil removal filters, water removal filters and dust removal filters (and an activated carbon absorber unit may also be used for oil vapour and hydrocarbon odour removal). Desiccant dryers are reliable, well-proven machines, however, it is important that they are correctly sized, controlled and maintained otherwise they have the potential to be high energy users, costing your business greatly. It is important not to oversize desiccant dryers, particularly the heatless models as the dryer has a fixed purge loss/purge cycle time. This means much energy is wasted if the demand is considerably less than the dryer design flow.

However, most dryers are now supplied with energy management control systems, such as dewpoint dependant switching. This make desiccant dryers more efficient and by using an energy management control system such as dew point switching, can lead to savings of up to 90%, depending on installation and usage. These models have built-in dewpoint switching or in-bed sensing devices that alter the level of regeneration according to the load level. Running costs will also rise significantly without proper operation and maintenance. Efficient operation of a desiccant dryer can also be hampered by:

- poor cooling of inlet air
- poor pre-filtration causing liquid phase water and oil carry over
- high peak loads causing desiccant bed fluidisation (twin tower type only)
- faulty change-over valves causing continuous purge
- faulty controls causing poor or no regeneration of individual towers
- desiccant contamination by oil

DEWPOINT ISSUES

No matter which configuration of dryers is chosen there are some common problems that impact on their efficiency costs. Often dryers have difficulty performing at expected efficiency due to poor ventilation and/or incorrect installation. Furthermore, inlet temperatures are frequently elevated above those allowed for in the system design causing poor dewpoints. A common cause of poor dewpoint is when different dryers are arranged in parallel-different makes of dryer, or dryers of different capacities creates unbalanced flow between dryers. The correct manifolding, piping and utilising either sonic nozzles or orifices can overcome this.

AIRLINE FILTERS

Air compressor systems need filters to create the different air quality levels required by different machines. Air filters can be located throughout the system and the number and type of filters will vary according to the quality of air required.

Compressed air has 3 contaminants: water, particulate and oil (if lubricated compressor) which all mix together to form an unwanted abrasive sludge. The same amount of water and dirt is generated whether a lubricated or non-lubricated compressor is used, therefore, both types of compressor need the same level of filtration. It also corroded piping systems which contributes to additional particulate (rust) contamination to downstream equipment and products. This all leads to decreased efficiency and increased energy consumption. It is, therefore, imperative that high efficiency filtration is selected which will maintain a low operation pressure differential whilst providing complete protection.

Air-filters can be divided into two categories: Pre-filters, which operate prior to compression and/or drying and After-filters which are put in place after the air is dried.

PRE-FILTERS

INLET FILTERS are provided to protect the compressor from incoming dirt. These filters use power to overcome pressure drop, however, this is taken into account in the compressor package performance figures. These filters will affect efficiency when they become dirty and can typically cause an increase in power consumption of 3%. Therefore, it is important to change the filters as recommended by the compressor service manual.

GENERAL PURPOSE FILTERS are usually installed between the after-cooler and drying process. Filters remove the contaminants from compressed air (water, particulate and oil) and are, therefore, required by non-lubricated and lubricated compressors.

HIGH EFFICIENCY OIL REMOVAL FILTERS provide air quality (technically oil-free) similar to that supplied by an 'oil-free' machine. It is able to remove almost all water and oil aerosol.

AFTER-FILTERS

DUST REMOVAL FILTERS are most commonly used with a desiccant dryer to remove the desiccant fines that are collected and carried downstream as the air encounters the drying process.

ACTIVATED CARBON FILTERS are required if the air is required for breathing; mixing with food, pharmaceutical products or other such duties. While other filters will remove water, liquid oil, oil aerosols, and dirt, only carbon filters will eliminate oil vapours and oil (and other-hydrocarbon) odours.

HOPKALITE FILTERS are required if the air is required for breathing and there is a risk of Carbon Monoxide being present in the system. These filters are used in conjunction with desiccant dryers, which with the use of the correct desiccant can reduce Carbon Dioxide levels and are required to keep the Hopkalite active.

STERILE FILTERS eliminate micro-organisms from compressed air and are used in processes which require the highest air quality. Sterilised filters are designed to be resterilised in place using steam.

POINT OF USE FILTERS are used to remove particles that have accumulated during distribution of compressed air. This contamination often occurs in piping. Particularly in ageing large systems where rust and pipe scale can collect. Activated Carbon filters are also used as point of use filters for the application as stated above.

Filters cause pressure drop and are often responsible for compressors generating at a pressure well above that required for the process. This occurs if filters are undersized, the wrong filters are being used for the job, too many filters are being used or filters are poorly maintained.

RECEIVER

Air receivers can be added to compressed air systems allowing the compressor capacity to be temporarily exceeded so that short-term demand spikes can be met. Receivers also create more stable pressure conditions. Working to dampen compressor pulsation, separate out particles and liquids, and make the compressed air system easier to control. Additionally, receivers act as a secondary cooling device. Installing a larger receiver tank to meet occasional peak demands can even allow a smaller compressor to be connected.

In most systems, the receiver will be located just before the dryer, as the receiver gives further cooling of the air prior to the dryer, and so reduces the amount of condensate the dryer has to deal with, increasing efficiency of the dryer. Alternatively, you may use multiple receivers, one prior to the dryer and others closer to the points of intermittent/high demand. Storage can be used to control peak demand periods in the system, to protect critical pressure applications from other events in the system or to control the rate of pressure drop in demand while supporting the speed of transmission response from supply. Often systems run a secondary compressor to cater for intermittent demand, compressor failure or short energy outages. The installation of air receivers may enable this compressor to be turned off.

On most installations, the receiver is fed from the after-cooler and further cooling will take place in the receiver. On installations where an after-cooler is not fitted, the receiver is where most condensed liquid is found. Therefore, the receiver needs to be fitted with drain traps, (preferably automatic) to remove the condensate and any carry-over solids such as dust, scale, carbon etc. The receiver should, wherever possible, be placed outside in a cool location, further reducing the temperature of the compressed air and increasing the amount of water and oil condensation. Additionally, some moisture can pass through the after-cooler condensate removal separator at high velocity. A receiver can help to trap this by creating a quiet zone where the turbulence is considerably reduced.